1 Determining mass action law

Given reaction:

$$aA + bB \leftrightarrow cC + dD$$

molar quantity	$\frac{\Delta G}{n} = \Delta \bar{G}$
partial molar quantity	$\frac{\partial \Delta G}{\partial n} = \mu$

Gibbs free energy:

$$dG = -SdT + VdP + \sum \mu_i dn_i$$

for a constant Temperature and Pressure process:

$$dG = \sum \mu_i dn_i$$

$$\mu_a dn_A + \mu_b dn_B = \mu_c dn_C + \mu_d dn_D$$

Assuming A, B, C, and D are all pure, $\Delta \bar{G} = \mu$

$$\bar{G}_i = \bar{G}_i^\circ + RT \ln P_i = \mu_i$$

$$n_A(G^\circ + RT\ln P)_A + n_B(G^\circ + RT\ln P)_B = n_C(G^\circ + RT\ln P)_C + n_D(G^\circ + RT\ln P)_D$$

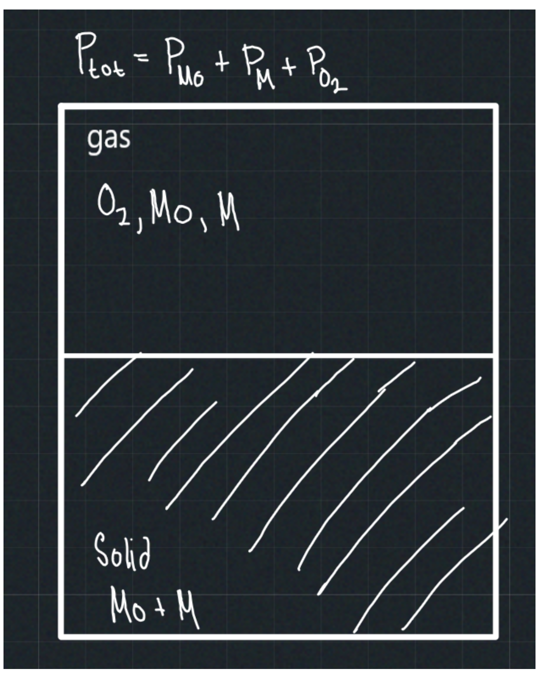
$$G^\circ_{\text{tot}} = \sum n_i G_i^\circ = n_A \bar{G}_A^\circ + n_B \bar{G}_B^\circ + n_C \bar{G}_C^\circ + n_D \bar{G}_D^\circ = (nRT\ln P)_A + (nRT\ln P)_B - (nRT\ln P)_C - (nRT\ln P)_D$$

$$\begin{aligned} G^\circ_{\text{tot}} &= RT\ln(\frac{P_A^{n_A} P_B^{n_B}}{P_C^{n_C} P_D^{n_D}}) \\ &= -RT\ln(\frac{P_C^{n_C} P_D^{n_D}}{P_A^{n_A} P_B^{n_B}}) \\ &= -RT\ln(k) \end{aligned}$$

2 Solid-gas reaction (Phase equilibria)

consider oxidation(two components):

$$M_{(s)} + 1/2 O_{2(g)} \leftrightarrow MO_{(s)}$$



$$
\begin{aligned}
\Delta \bar{G}_i &= \Delta G_i^\circ + RT \ln P_i \\
\mu_i &= \mu_i^\circ + RT \ln P_i \\
G_{\mathrm{tot}}^\circ &= \sum n_i G_i^\circ = \sum \mu_i d n_i =
\end{aligned}
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